

KCT CRM

Complete Functional, Workflow & Product Roadmap Blueprint

Kiwi Cloud Tech - Internal Enterprise CRM Platform

Document Version:	Production Baseline (v1.0) + Extended Product Roadmap (v2 / v3)	Date of Issue:	September 23, 2026
System Identity:	Kiwi Cloud Tech Internal CRM (Repository: D:\kct\crm)	Target Audience:	Core Engineering, Product Teams, Operations & Leadership
Classification:	Internal Company Confidential - Proprietary Operating Asset	System Nature:	Unified Operational Nervous System (Lead to Cash & Delivery)

CRITICAL MANDATE & ARCHITECTURAL SCOPE

SYSTEM BOUNDARY NOTICE: The KCT CRM is exclusively an **internal corporate CRM** designed to manage customer acquisition, client companies, contracts, multi-track delivery, QA verification, customer service, and double-entry accounting operations. **The CRM is strictly NOT a student-facing LMS (Learning Management System) or student portal.** It contains no student course players, exam proctoring engines, or student gradebooks. All training/edtech records exist strictly as commercial client contracts, institutional accounts, and enrolled pipeline opportunities.

PURPOSE & AUTHORITATIVE CHARTER

EXECUTIVE BLUEPRINT SUMMARY:

This blueprint establishes the authoritative technical, operational, and functional baseline for the **Kiwi Cloud Tech CRM**. It documents the complete end-to-end commercial lifecycle across three distinct operating verticals: **EdTech Commercial, IT Services & Engineering**, and **Talent / Staff Augmentation**. The system unifies demand generation, pipeline velocity, quote-to-order conversions, milestone-driven delivery, defect-gated QA, SLA-bound support, and general ledger accounting into a single, highly auditable Company 360 cockpit. This document serves as the permanent contract for feature development, engineering discipline, security standards, and the phased rollout of future v2/v3 capabilities.

Section Group	Covered Functional Modules	Strategic Focus
I. Foundations & Strategy	Sections 1 - 4: Executive Overview, Objectives, Lifecycle Flow, 3 Pipelines	Business context, multi-track pipeline models
II. Core Operations	Sections 5 - 11: CRM Core, Company 360, Leads, Activities, Opportunities, Quotes, Orders	Demand management, Company 360, deal velocity
III. Delivery & Service	Sections 12 - 16: Project Delivery, QA Gating, Service & Support, SLA, Accounting	Execution quality, defect gates, financial ledger
IV. Intelligence & Comms	Sections 17 - 22: Comms Architecture, Dashboard, Search/Reports, RBAC, Security, CSV	Observability, RBAC security, data portability
V. Future Roadmap	Sections 23 - 30: Automation v2, Notifications, Lifecycle, Docs, AI v3, Mobile, Roadmaps	Next-gen automation, AI copilot, native mobile
VI. Architecture & Governance	Sections 31 - 36: Master Flow, Current State, 15 Principles, SDLC Rules, Priorities, Arch	Engineering standards, deployment priorities

SECTION 01

EXECUTIVE OVERVIEW

What KCT CRM Is: KCT CRM is the proprietary, full-stack customer relationship, commercial execution, and operational lifecycle management system built specifically for **Kiwi Cloud Tech**. It serves as the single operational source of truth connecting initial market demand to financial revenue recognition and post-delivery client retention.

Why KCT CRM Exists: Prior to centralizing operations on KCT CRM, multi-vertical business interactions were fragmented across disparate spreadsheets, local email folders, unlinked WhatsApp chats, and disconnected task trackers. KCT CRM eliminates operational blind spots, guarantees follow-up discipline, enforces quality gates before software delivery, ensures SLA compliance for customer issues, and links every operational rupee to an auditable accounting invoice.

Who Uses It (User Personas):

- * **Executive Leadership & Management:** Reviews real-time pipeline velocity, financial margins, SLA breach alerts, and company-wide KPI dashboards.
- * **Sales Managers & Executives:** Manages multi-channel lead ingestion, schedules mandatory follow-ups, drives opportunity stages, and generates line-item quotations.
- * **Project Managers & Technical Leads:** Converts confirmed sales orders into milestone-governed delivery projects, breaks down engineering tasks, and monitors delivery burn rates.
- * **Quality Assurance (QA) Engineers:** Builds test suites, executes test runs, logs defects, and enforces the mandatory zero-critical/high defect delivery gate.
- * **Customer Support & Service Representatives:** Resolves client tickets, responds within contractual SLA windows, and collaborates via private internal notes.
- * **Finance Operations & Accountants:** Generates tax-compliant invoices, applies customer payments, reviews double-entry journal entries, and manages accounts receivable.

Core Operational Taxonomy (Module Differences):

The architecture strictly decouples operational layers to avoid monolithic entanglement: **CRM Core** governs identity, companies, contacts, and universal audit logging; **Sales** focuses on lead velocity, opportunity conversion, and quotation acceptance; **Delivery** tracks project milestones and development sprints; **QA** acts as the quality firewall blocking unverified releases; **Service** ensures post-release customer SLA resolution; **Accounting** guarantees double-entry ledger integrity; **Communication** provides provider-neutral omni-channel threading; and **AI** acts purely as an augmenting intelligence layer without displacing transactional logic.

EXPLICIT BOUNDARY REITERATION

NON-LMS DECLARATION: The KCT CRM is explicitly an internal enterprise tool. It does NOT host student video courses, quizzes, certificates, or student community boards. EdTech programs are modeled strictly from a business perspective as corporate training contracts, institutional partnerships, or commercial student enrollments.

SECTION 02

BUSINESS OBJECTIVES

Strategic Objective	Operational Capability	Core Business Metric / Impact
1. Centralize Company Data	Unified Company 360 cockpit aggregating all contracts, tickets, and activities.	Elimination of data silos; 100% account history visibility.
2. Lead Capture & Assignment	Multi-channel intake (16 sources) with instant owner routing.	Sub-15 minute initial response time for inbound prospects.
3. Follow-Up Governance	Mandatory Next-Action and Next-Follow-up enforcement on all active leads.	Zero lead drop-offs; 35% higher lead-to-opportunity conversion.
4. Multi-Pipeline Sales	Dedicated stages for EdTech, IT Services, and Staff Augmentation.	Accurate deal velocity tracking and pipeline weighted value.
5. Quotation Generation	Standardized product/service line items with automatic taxes and discounts.	Quote turnaround reduced from 24 hours to 10 minutes.
6. Delivery Governance	Direct sales-order-to-project handover with milestone tracking.	On-time delivery tracking; seamless sales-to-engineering handoff.
7. Quality Gatekeeping	Integrated bug tracker with mandatory release blocking on Critical/High bugs.	Zero severe defects released to production clients.
8. SLA-Backed Support	Automated response and resolution timers with status-aware pausing.	>95% contractual SLA compliance across all client accounts.

9. Financial Integrity	Integrated invoicing, payment tracking, and double-entry journal sync.	Zero unbilled deliveries; reduction of Days Sales Outstanding (DSO).
10. Omni-Channel Trail	Centralized logging for emails, calls, WhatsApp, and internal notes.	Full accountability across team handoffs and client disputes.
11. Executive Visibility	Real-time cross-departmental dashboards and customizable CSV exports.	Instant operational decision-making without manual reporting.
12. Repetitive Automation	Automated task creation, status triggers, and SLA escalations.	30% reduction in administrative overhead for managers.
13. Augmented Intelligence	AI summarization, response suggestions, and predictive forecasting (v3).	Enhanced agent productivity and accurate revenue forecasting.

SECTION 03

OVERALL END-TO-END BUSINESS FLOW

KCT CRM connects every commercial touchpoint across the entire client lifecycle. The following sequential workflow illustrates how an initial market touchpoint evolves into long-term enterprise revenue:

1. Lead Generation	2. Lead Qualification	3. Sales Pipeline	4. Opportunity	5. Quotation / Proposal
1. Negotiation	2. Deal Won	3. Sales Order / Contract	4. Project Delivery	5. QA Verification
1. Delivery Sign-off	2. Customer Support	3. Invoice Generated	4. Payment Collected	5. Retention / Referral

Detailed Phase-by-Phase Breakdown:

- * **1. Lead Generation & Capture:** Inbound prospects arrive via web forms, social ads, partner referrals, or direct outreach, immediately creating a central Lead record.
- * **2. Qualification & Routing:** Leads are evaluated against ICP (Ideal Customer Profile) criteria and routed to the appropriate specialist sales owner.
- * **3. Opportunity & Pipeline Structuring:** Qualified leads are converted into Opportunities and mapped to one of the 3 dedicated business pipelines.
- * **4. Quoting & Negotiation:** Official line-item quotations with tax and validity terms are generated. Revisions are tracked until client consensus is reached.
- * **5. Won & Commercial Commitment:** Upon acceptance, the deal marks as Closed-Won, triggering legal Contract generation and binding Sales Order creation.
- * **6. Project Kickoff & Engineering:** Delivery teams receive the SOW (Statement of Work), establishing milestones, deliverables, and engineering sprints.
- * **7. QA Defect Gating:** The QA team executes test cases. All Critical/High bugs must be resolved before client release authorization.
- * **8. Delivery Sign-off & Invoicing:** Client accepts delivery. Finance automatically generates the tax invoice with specified credit terms.
- * **9. Support & SLA Resolution:** Ongoing operational support is managed via SLA-governed tickets with complete history preserved in Company 360.
- * **10. Payment, Retention & Expansion:** Collections are reconciled into the accounting ledger, initiating the renewal and referral pipeline.

SECTION 04

THREE DEDICATED BUSINESS PIPELINES

Architectural Foundation: Kiwi Cloud Tech operates across three radically different commercial models. Forcing these diverse business motions into a single generic status model (such as 'New -> Open -> Closed') destroys visibility, muddles sales velocity metrics, and frustrates account executives. KCT CRM enforces **three independent pipeline state machines**:

PIPELINE 1 - EDTECH COMMERCIAL (Institutional & Training Programs)

1. Lead	2. Qualified	3. Counselling	4. Demo Session	5. Payment	6. Enrolled	7. Completed	8. Referral
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Characteristics: High lead volume, short cycle (3-14 days), counselling/advising-centric, direct tuition/fee collection, and post-completion alumni referral loops.

PIPELINE 2 - IT SERVICES (Custom Software, Cloud & DevOps)

1. Prospect	2. Contacted	3. Discovery	4. Requirement	5. Proposal	6. Negotiation	7. Won	8. Delivery	9. Retainer
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Characteristics: High deal value, multi-week consultative cycle (30-90 days), technical architecture discovery, custom SOWs, formal milestone delivery sign-offs, and multi-year retainer conversions.

PIPELINE 3 - TALENT / OUTSOURCING (Staff Augmentation & Recruiting)

1. Company Prospect	2. Requirement	3. Candidate Search	4. Profiles Shared	5. Client Interview	6. Selected	7. Joined	8. Invoice
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Characteristics: Highly time-sensitive, turnaround measured in hours/days, candidate profile match matching, multi-stage technical interviews, placement risk buffers until joining date, and placement fee billing.

STRICT ISOLATION POLICY

DATA INTEGRITY RULE: Pipeline stages are non-interchangeable. An EdTech prospect cannot have an 'Interview' stage, and a Staff Augmentation deal cannot have a 'Demo Session' stage. Each pipeline enforces its own stage validations, probability weightings, and SLA metrics.

SECTION 05 CRM CORE FUNCTIONALITY		
Core Module	Functional Purpose	Key Data Entities & Capabilities
Authentication & Auth	Identity, session security, password policies.	Argon2/bcrypt hashing, JWT access tokens, secure HTTP-only refresh tokens.
Users & RBAC	Team identity, department routing, role permissions.	User profiles, role assignments (Admin, Sales, QA, PM, Support, Finance), action ACLs.
Companies	Primary master organizational entity for clients/vendors.	Name, domain, GST/tax ID, industry, billing/shipping address, tier, Company 360 link.
Contacts	Individual business stakeholders linked to companies.	First/last name, work email, direct mobile, designation, decision-maker flag, social URLs.
Leads	Unqualified market demand prior to commercial qualification.	Source, status, owner, business segment, contact details, mandatory next-action date.
Activities	Universal audit trail of all customer interactions.	Phone calls, outgoing emails, meeting notes, internal observations, file attachments.
Follow-ups	Actionable commitments with strict calendar deadlines.	Scheduled timestamp, assigned owner, completion status, priority, auto-overdue tracking.
Tasks	General operational and administrative assignments.	Due date, status (Pending, In Progress, Completed), project/ticket association, checklists.
Pipelines & Deals	Commercial opportunity tracking across 3 distinct funnels.	Pipeline selector, custom stages, expected deal value, probability, expected close date.
Dashboard & Analytics	Real-time management observability and KPI cockpits.	Pipeline value, won revenue, team activity metrics, SLA breaches, outstanding cash.
Global Search	Universal indexing across all major business entities.	Sub-second multi-entity lookup (Companies, Contacts, Leads, Invoices, Projects, Tickets).
CSV Portability	Bulk data ingestion and standardized extraction.	Multi-entity import with duplicate checking, preview validation, and full CSV export.
Audit Logging	Immutable compliance ledger for sensitive operations.	Actor ID, action type, entity name, target ID, client IP, timestamp, JSON delta changes.

SECTION 06COMPANY 360 - THE MASTER ORGANIZATIONAL ENTITY

The Single Source of Client Truth: In KCT CRM, the **Company** entity is the foundational root to which all other operational and financial records anchor. No client contract, project, or invoice can exist in an isolated vacuum. When an account executive or executive opens a Company record, the **Company 360 Cockpit** aggregates every historic and active interaction:

Company (Root Client / Organization Entity)

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Contacts (Authorized Stakeholders & Decision Makers)

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Leads (Inbound Inquiries & Cross-sell Opportunities)

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Opportunities & Deals (Commercial Pipeline across EdTech, IT, Talent)

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Activities (Complete Historic Timeline of Calls, Meetings, Notes)

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Follow-ups (Scheduled Future Commitments & Action Items)

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Quotations (Formal Price Proposals & Scope Estimates)

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Sales Orders (Binding Commercial Commitments)

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Contracts (Legal MSAs, SOWs, Retainer Agreements & Renewal Dates)

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Projects (Engineering Sprints, Milestones & Deliverables)

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QA Defect Logs (Bugs, Test Suites & Quality Sign-offs)

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Service Tickets (Support Inquiries, SLAs & Incident Resolutions)

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Invoices (Tax-Compliant Invoices, Credit Notes & Payment Schedules)

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Payments (Collected Receipts & General Ledger Reconciliation)

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Communication History (Omni-channel Email & WhatsApp Threads)

Key Operational Advantage: When customer support receives an escalation, they immediately see if the client has an active critical invoice pending or an ongoing software milestone delivery. When sales pitches a renewal, they inspect unresolved support tickets. Siloed departmental blindness is completely eliminated.

SECTION 07

LEAD MANAGEMENT WORKFLOW

1. Lead Creation	2. Auto / Manual Assignment	3. Qualification	4. Activity Logged	5. Follow-up Scheduled	6. Opp Conversion	7. Closed / Lost
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The 16 Multi-Channel Lead Sources: To accurately measure marketing ROI and acquisition channel velocity, every lead entering KCT CRM must be tagged with one of the 16 recognized source categories:

Source Track A	Source Track B	Source Track C	Source Track D
1. Website (Inbound Form)	5. LinkedIn Outreach	9. Webinar Attendee	13. Cold Calling
2. Google Search Ads	6. YouTube Channel	10. Technical Workshop	14. Business Partner
3. Instagram Campaign	7. WhatsApp Inbound	11. Client Referral	15. Existing Client Up-sell
4. Facebook Campaign	8. College / Campus Drive	12. Cold Email Campaign	16. Organic Web Search

MANDATORY NEXT-ACTION DISCIPLINE

CRITICAL LEAD GOVERNANCE RULE: Every active lead in KCT CRM MUST have an explicitly defined **Next Action** and an assigned **Next Follow-Up Date & Time**. Leads cannot sit in an indefinite 'In Progress' state without a calendar-scheduled touchpoint. Leads violating this rule are automatically flagged in the manager's Today Work Center as 'Abandoned Leads'.

Complete Lead Record Anatomy:

- * **Lead Owner:** Assigned sales executive responsible for lead progression.
- * **Priority:** Categorized as Low, Medium, High, or Urgent based on timeline and budget.
- * **Source:** One of the 16 standardized marketing and outreach channels.
- * **Business Segment:** EdTech, IT Services, or Talent Staffing.
- * **Target Pipeline:** Specific pipeline funnel the lead will enter upon qualification.
- * **Stage:** New, Contacted, Qualified, Unqualified, or Nurturing.
- * **Contact Details:** Name, direct telephone, verified work email, location, and timezone.
- * **Company Linkage:** Associated Company entity (auto-matched by email domain or manually linked).
- * **Next Action & Follow-up:** Mandatory text description of agreed commitment and exact future timestamp.
- * **Full Activity History:** Chronological audit stream of every call, email, and internal note.

SECTION 08

ACTIVITIES & FOLLOW-UPS (TODAY'S WORK CENTER)

Chronological Customer Timeline: Client interactions are logged as immutable Activity records. Every logged activity immediately updates the Company, Contact, and Lead timeline, providing transparent handoffs across team members.

6 Core Interaction Types: 1. **Phone Calls** (duration, outcome, discussion points), 2. **Emails** (sent/received body, attachments, delivery status), 3. **Client Meetings** (agenda, attendee list, meeting minutes), 4. **Internal Notes** (private team observations hidden from external clients), 5. **Follow-Ups** (commitments requiring future outreach), and 6. **Task Completions** (deliverable checklists and administrative items).

Today's Work Center - The Sales Execution Cockpit:

To maximize daily execution velocity, sales reps begin every morning inside the **Today's Work Center**. This dynamic operational view prioritizes immediate revenue risks into 5 actionable buckets:

Work Center Bucket	Operational Criteria	Required Action
1. Follow-ups Due Today	Follow-ups scheduled for the current calendar date.	Execute call/meeting, record outcome, schedule next action.
2. Overdue Follow-ups	Past-due follow-ups where scheduled date < today.	Immediate high-priority recovery touchpoint or manager re-assignment.
3. Upcoming Follow-ups	Follow-ups scheduled across the next 3 to 7 business days.	Pre-meeting preparation, technical collateral compilation.
4. Unassigned Leads	Newly ingested inbound leads without an allocated owner.	Sales manager triage and round-robin allocation within 15 minutes.
5. Leads Without Next Action	Active leads lacking a scheduled follow-up date/time.	System compliance violation; rep must define next commitment.

SECTION 09

SALES / OPPORTUNITY MANAGEMENT

1. Lead Converted	2. Opportunity Created	3. Pipeline Assigned	4. Stage Advanced	5. Value & Close Date	6. Quotation Attached	7. Won / Lost Closing
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Opportunity Data Architecture & Weighted Pipeline: When a lead demonstrates budget, authority, need, and timeline (BANT), it converts into an Opportunity. Each opportunity is governed by:

- * **Opportunity Name:** Standardized format: [Company Name] - [Service / Product] - [Fiscal Quarter].
- * **Pipeline Selection:** Strictly routed to EdTech Commercial, IT Services, or Talent Outsourcing.
- * **Stage Progression:** Sequential stage advancement enforcing milestone exit criteria.
- * **Expected Contract Value:** Total gross commercial commitment in INR / USD.
- * **Probability Weighting (%):** Automatically populated based on current stage (e.g., Discovery = 20%, Proposal = 60%, Negotiation = 80%).
- * **Weighted Pipeline Calculation:** Expected Value × Stage Probability used for cashflow forecasting.
- * **Expected Close Date:** Target closing date with automatic alerts if deal slips past month-end.
- * **Product & Service Catalog Linkage:** Multi-line selection of SKUs, hourly rates, course fees, or placement tiers.
- * **Won / Lost Attribution:** On closing, reps must record Won Revenue or select mandatory Lost Reasons (Price, Competitor, Stalled, Scope).

SECTION 10

QUOTATIONS & PRICE PROPOSALS

Line-Item Quotation Architecture: Quotations are formal financial instruments generated directly from an Opportunity. They establish exact price transparency and contractual clarity before work commences.

Quotation Field / Section	Data Specification & Business Rules	Operational Impact
Header & Client Details	Auto-populated from Company and Contact records (Billing/Shipping Address, GSTIN).	Eliminates manual re-typing and invoicing errors.
Product / Service Line Items	Item name, description, SKU, quantity, unit list price, and unit of measure.	Ensures consistent catalog pricing across all sales reps.

Discounts & Deductions	Line-item percentage discount or overall commercial discount.	Discounts >15% require Sales Director approval (v2).
Taxation Engine	Automatic calculation of CGST, SGST, IGST (standard 18%) or international 0% export tax.	Guarantees GST compliance and automated tax ledger sync.
Validity & Commercial Terms	Quotation expiration date (default: 30 days) and explicit payment milestone terms.	Protects margins against inflationary wage/resource changes.
Scope Assumptions & Notes	Specific inclusions, exclusions, client prerequisites, and warranty periods.	Prevents scope creep during technical project delivery.

Future v2 Quotation Roadmap: 1. One-click dynamic PDF generation with Kiwi Cloud Tech corporate branding, 2. Multi-tier approval workflows based on deal discount thresholds, 3. Customer digital acceptance portal with timestamped audit logs, 4. Full versioning (Quote v1, v2, v3) preserving quote revision history, and 5. Integrated cryptographic e-signature sign-off.

SECTION 11

SALES ORDERS & CONTRACTS

1. Quotation Accepted	2. Sales Order Generated	3. Contract Drafted	4. Legal Execution	5. Project Delivery Kickoff
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- Commercial Execution Bridge:** A quotation is an offer; a **Sales Order** and **Contract** represent binding commercial commitments. When an opportunity is marked Closed-Won, KCT CRM transitions the transaction into an operational order:
- * **Sales Order (SO):** Captures accepted line items, committed delivery timelines, and authorized client PO numbers. Serves as the operational trigger for engineering and finance.
 - * **Contract Entity:** Governs the legal relationship, Master Services Agreement (MSA), Statement of Work (SOW), or Retainer Contract.
 - * **Contract Numbering:** Automated sequence format: KCT-CON-[YEAR]-[SEQ].
 - * **Contract Term & Dates:** Effective Start Date, Contractual End Date, and Notice Period for Termination.
 - * **Commercial Values:** Total Contract Value (TCV) and Annualized Contract Value (ACV).
 - * **Billing Frequency:** Milestone-based (e.g. 40-30-30), Monthly Recurring Retainer, or Net-30 Upfront.
 - * **Renewal Notification Date:** Automated trigger set 45 days prior to contract expiration.
 - * **Contract Status:** DRAFT, ACTIVE, EXPIRING_SOON, RENEWED, or TERMINATED.

AUTOMATED REVENUE RETENTION
RENEWAL GOVERNANCE (Future v2): The CRM automatically creates a high-priority Renewal Task in the Account Owner's queue exactly 45 days prior to contract expiration, safeguarding Kiwi Cloud Tech's recurring retainer revenue.

SECTION 12

PROJECT DELIVERY & ENGINEERING GOVERNANCE

1. Sales Order	2. Project Created	3. Team Assigned	4. Milestones Set	5. Task Sprints	6. Dev / Build	7. QA Gate	8. Delivery Acceptance
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- Bridging Sales to Production:** Too many CRMs stop when the deal closes, leaving delivery isolated in external tools. KCT CRM includes native project delivery management linked directly to the originating Sales Order and Company 360 profile:
- * **Direct Project Spawning:** Confirmed Sales Orders automatically instantiate a Project record with scope pre-populated from the quotation.
 - * **Project Team Allocation:** Assignment of Project Manager (PM), Technical Architects, Senior Developers, and dedicated QA Leads.
 - * **Milestone Tracking:** High-level commercial delivery gates (e.g. Architecture Sign-off, Alpha Release, Beta Release, Production Deployment).
 - * **Granular Task Management:** Engineering tasks with priority, estimated hours, actual hours, completion state, and blockers.
 - * **Budget & Burn Rate:** Real-time tracking of consumed engineering hours against the commercial quotation budget.
 - * **Client Transparency:** Ability to export milestone progress reports for client steering committee reviews.

SECTION 13

QUALITY ASSURANCE (QA) & CRITICAL BUG DELIVERY GATE

1. Test Suite Created	2. Test Cases Authored	3. Test Execution	4. Bug Logged	5. Developer Fix	6. QA Retest	7. Verified Pass
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- Integrated QA Engineering:** Quality is not an afterthought in KCT CRM. The system provides complete test management and defect tracking directly within the delivery workspace:
- Project Release Lifecycle States:** Projects advance through 6 sequential operational phases: PLANNED -> ACTIVE -> IN_QA -> READY_FOR_DELIVERY -> DELIVERED -> COMPLETED.

MANDATORY QUALITY RELEASE FIREWALL
CRITICAL QUALITY GATE RULE: The KCT CRM backend strictly enforces that a project CANNOT transition to **READY_FOR_DELIVERY** if there are any unresolved bugs marked with severity **CRITICAL** or **HIGH**. If a project manager attempts to authorize delivery while severe defects remain open, the system throws an explicit validation error: 'Cannot mark project READY_FOR_DELIVERY: Unresolved Critical/High severity bugs exist'. This architectural firewall protects Kiwi Cloud Tech's brand reputation and prevents buggy code from reaching production clients.

Bug Tracking Anatomy: Every defect record tracks Bug ID (e.g. BUG-2026-0001), Project ID, Title, Steps to Reproduce, Severity (Critical, High, Medium, Low), Priority, Assigned Developer, Testing Environment, File Attachments (stack traces, screenshots), and Resolution Status (Open, In Progress, Resolved, Verified, Closed).

SECTION 14

CUSTOMER SERVICE & SUPPORT WORKFLOW

1. NEW	2. OPEN	3. ASSIGNED	4. IN_PROGRESS	5. WAITING_FOR_CUSTOMER	6. RESOLVED	7. CLOSED
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- Post-Delivery Operational Support:** Client issues, service requests, and maintenance inquiries are handled through the native Service Desk module. Every ticket is linked to its master Company record and tracks complete status history:
- * **Standard Ticket Lifecycle:** Follows strict stage progression from initial ingestion to verified resolution.
 - * **Reopen State Machine:** If a customer responds to a closed ticket with unresolved concerns, the system automatically transitions: CLOSED -> REOPENED -> IN_PROGRESS, alerting the original support agent.
 - * **Categories & Subcategories:** Technical Bug, Configuration Request, Billing Inquiry, Feature Request, or Outage.
 - * **Internal Notes vs Public Comments:** Agents can post private internal notes (collaboration between developers and support) completely hidden from the customer's view.
 - * **Attachment Security:** Secure multi-file upload for error logs, diagnostics, and screenshots with path traversal protection.
 - * **Status Audit Trail:** Every status transition logs the actor, timestamp, and duration spent in each state.
 - * **Multi-tier Escalation:** Unresolved tickets automatically escalate from L1 Helpdesk to L2 Support to L3 Engineering.

SECTION 15

SLA MANAGEMENT & ACTIVE RESPONSE PAUSING

1. Ticket Ingested	2. SLA Policy Applied	3. Response Deadline Set	4. Resolution Deadline Set	5. Monitoring & Warning	6. Escalation	7. Resolved
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Contractual Service Level Agreements (SLAs): To maintain enterprise credibility, KCT CRM monitors response deadlines (First Agent Reply) and resolution deadlines based on ticket priority:

Severity Level	First Response SLA	Full Resolution SLA	Escalation Trigger
CRITICAL (P1 - Outage / System Down)	15 Minutes	4 Hours	Immediate SMS / Phone Alert to Technical Lead
HIGH (P2 - Core Feature Blocked)	1 Hour	12 Hours	Notification to Engineering Manager at 75% timer
MEDIUM (P3 - Degraded Performance)	4 Hours	24 Hours	Dashboard warning at 80% timer elapsed
LOW (P4 - Minor Inconvenience / Question)	8 Hours	72 Hours	Daily operational digest report

INTELLIGENT SLA TIMER GOVERNANCE
ACTIVE SLA PAUSE & RESUME MECHANISM: When an agent requests additional logs, reproduction steps, or credentials from a client, the ticket status changes to **WAITING_FOR_CUSTOMER**.
The KCT CRM SLA engine **automatically pauses the SLA resolution countdown timer** during this state. Support agents are not penalized for client-side response delays. As soon as the customer submits their reply, the ticket shifts back to **IN_PROGRESS**, and the SLA timer resumes from where it left off.

Future SLA Capabilities (v2): Automated proactive warning notifications (yellow alert at 75% elapsed), automatic re-assignment to secondary on-call engineers, and monthly executive SLA compliance scorecards.

SECTION 16

DOUBLE-ENTRY ACCOUNTING & COMMERCIAL FLOW

The CRM-to-Ledger Bridge: KCT CRM features a complete financial ledger foundation built directly into the core schema. This bridges customer delivery directly to cash collection:

1. Sales Order	2. Tax Invoice Generated	3. Payment Collected	4. General Ledger Journal Entry	5. Revenue Recognized
Accounting Entity / Subsystem	Technical Architecture		Operational CRM Integration	
Chart of Accounts	Structured 5-category ledger: Assets, Liabilities, Equity, Revenue, and Expenses.		Standard account numbering supporting multi-entity reporting.	

Tax Invoices	GST-compliant invoices with unique sequential numbering (e.g. INV-2026-0001).	Generated with one click from confirmed Sales Orders.
Customer Payments	Receipt tracking supporting Bank Wire, UPI, Credit Card, and Cheque.	Applies partial or full payments directly against open invoices.
Bills & Vendor Expenses	Accounts Payable (AP) ledger for contractor fees and infrastructure costs.	Links third-party costs directly to specific client project margins.
Journal Entries	Strict double-entry validation (total debits must exactly equal total credits).	Automatically generated upon invoice issue and payment receipt.
Bank Reconciliation	Bank statement transaction matching against recorded receipts.	Ensures real cash balance matches recorded CRM ledger balance.
Fiscal Periods & Tax	Monthly and annual accounting periods with GST tax liability reporting.	Provides instant tax export for statutory CA/audit filings.

SEPARATION OF CONCERNS
UX DESIGN PRINCIPLE: While KCT CRM possesses deep double-entry ERP accounting capabilities, **accounting screens do NOT clutter or dominate the daily sales and service CRM experience.** Sales reps only see clean Invoice status (Draft, Sent, Paid, Overdue) and Payment receipts on their Company and Opportunity screens. Deep journal entries and chart-of-accounts configurations remain isolated within dedicated Finance roles.

SECTION 17

COMMUNICATIONS ARCHITECTURE & INTEGRATION STATUS

Omni-Channel Engagement Foundation:

KCT CRM is engineered to centralize all client communications into a single unified thread under the Contact and Company profile:

EMAIL CHANNEL:

CRM Interface -> Company Mailbox (Hostinger SMTP/IMAP) -> Customer -> Reply Threading -> Activity Log

CALLING CHANNEL:

CRM Interface -> Telephony Gateway -> Dedicated Business Number -> Customer -> Audio Log -> Follow-up

WHATSAPP CHANNEL:

CRM Interface -> WhatsApp Cloud API / BSP -> Direct Customer Chat -> Conversation Thread -> Activity Log

INTENTIONAL STUB / PROVIDER-NEUTRAL STATUS

CRITICAL ARCHITECTURAL STATUS (CURRENT PRODUCTION BASELINE):

In the current baseline codebase, **Email and Telephony providers are intentionally DISABLED and decoupled from live external production gateways.** The application utilizes clean **provider-neutral abstraction layers** (e.g. MAIL_PROVIDER=none, TELEPHONY_ENABLED=false).

This design choice ensures complete local portability, prevents accidental credential exposure during testing, and guarantees that developers cannot trigger real external emails or phone charges during development and regression test runs.

Live Hostinger SMTP/IMAP and Cloud Telephony gateways are scheduled for Phase 3 activation after full domain verification and production hardening.

SECTION 18

DASHBOARD KPIS & EXECUTIVE OBSERVABILITY

Real-Time Operational Cockpit:

The KCT CRM dashboard delivers instantaneous visibility across 5 key operational domains, eliminating manual spreadsheets and end-of-month reporting scrambles:

Domain Panel	Tracked Operational KPIs	Managerial Purpose
1. Sales & Pipeline	New Leads (Today/Week), Qualified Leads, Active Opportunities, Pipeline Gross Value, Won Revenue, Lost Value.	Assesses acquisition funnel health and forecasted sales conversions.
2. Activity & Productivity	Calls Logged, Emails Dispatched, Client Meetings, Follow-ups Completed Today, Overdue Follow-up Backlog.	Tracks sales rep outreach cadence and identifies follow-up bottlenecks.
3. Delivery & QA	Active Client Projects, Projects Currently in QA, Blocked / At-Risk Projects, Open Defect Backlog (Critical/High).	Guarantees on-time milestone delivery and enforces quality release gates.
4. Service & Support	Open Tickets, Critical Incident Count, Active SLA Breaches, Average Resolution Time (MTTR), Customer Satisfaction.	Monitors client satisfaction and flags contractual SLA exposure.
5. Financial & Revenue	Billed Invoices Count, Total Outstanding Receivables (AR), Collected Payments This Month, Aging Receivables (>60 days).	Provides cashflow visibility and drives timely collection follow-ups.

SECTION 19

SEARCH & MULTI-ENTITY REPORTING

Universal Multi-Entity Search:

A global search bar at the top of the interface enables instant sub-second lookup across all primary entities: **Companies** (by name, domain, tax ID), **Contacts** (by name, email, phone), **Leads** (by title, company), **Opportunities** (by deal name), **Projects** (by code, title), **Tickets** (by ticket ID, subject), **Invoices** (by invoice number), **Contracts** (by contract ID), and **Activities**.

Built-In Reporting Engine:

1. **Sales Funnel Report** (stage-by-stage drop-off conversion rates), 2. **Pipeline Forecast** (expected close dates vs weighted revenue), 3. **Team Activity Leaderboard** (outreach calls, emails, and meetings per rep), 4. **Project Health Matrix** (milestone completion % vs allocated budget), 5. **QA Defect Density** (bugs per project, time to fix, severity distribution), 6. **Service SLA Report** (compliance % by priority, breach log), 7. **Accounts Receivable Aging** (current, 30 days, 60 days, 90+ days overdue), and 8. **Channel Acquisition ROI** (revenue won per lead source).

Future Advanced Analytics (v2 / v3):

Sales velocity modeling, customer health scoring algorithms, predictive churn warnings, and dynamic project margin profitability analysis.

SECTION 20

ROLE-BASED ACCESS CONTROL (RBAC)

Granular Multi-Tier Authorization: KCT CRM enforces strict role-based access control. Permissions govern both UI visibility and backend API endpoint access:

System Role	Primary Scope & Functional Responsibilities	Access Restrictions & Security Gates
Super Admin	Total system configuration, user provisioning, global overrides.	Full access; all administrative overrides logged to immutable audit trail.
Executive Management	High-level dashboards, cross-departmental reports, financial margins.	Read-only access to engineering tasks; full visibility into financial cashflow.
Sales Manager	Team lead allocation, pipeline oversight, quota tracking, quote approvals.	Restricted from backend server configuration and direct journal adjustments.
Sales Executive	Lead qualification, daily follow-ups, opportunity progression, quote creation.	Access limited to owned leads/deals or team-shared territory accounts.
Project Manager	Project creation, milestone planning, resource task allocation, release prep.	Cannot alter financial invoices or modify system authentication settings.
Developer / Engineer	Task execution, time tracking, bug fixes, pull request linkages.	Restricted from commercial pricing, billing invoices, and client contact lists.
QA Lead / Engineer	Test suite creation, bug logging, verification, delivery gate sign-off.	Has absolute authority to block READY_FOR_DELIVERY; cannot edit invoices.
Service Support Rep	Ticket triage, customer communication, SLA compliance, knowledge base.	Cannot access company financial records, sales quotas, or development tasks.
Finance Operations	Invoice issuance, payment reconciliation, credit notes, general ledger.	Restricted from software release gates, QA suites, and code repos.

SECTION 21

SECURITY ARCHITECTURE & AUDIT INTEGRITY

Defense-in-Depth Security Architecture:

Built to enterprise security standards, KCT CRM implements multi-layer protection across all data flows:

* Authentication Security:

Industry-standard password hashing using Argon2 and bcrypt. Multi-factor authentication (MFA) foundation.

* Stateless JWT Tokens:

Short-lived access tokens (15-60 min) paired with rotating refresh tokens stored in secure, HTTP-only cookies.

* Immutable Audit Trail:

Dedicated audit_logs table tracking actor ID, action verb, target table, record ID, client IP address, user agent, and JSON delta changes.

* File Security & Validation:

Uploaded attachments are strictly validated for MIME type, file extension whitelist, and maximum size (10 MB). Filenames are sanitized to prevent path traversal.

* HTTPS / TLS Enforcement:

Nginx reverse proxy configuration strictly enforces modern TLS 1.2/1.3 ciphers and HSTS headers.

* Rate Limiting (DDoS Protection):

Endpoint-specific throttling using SlowAPI/Redis (e.g. Login: 10/min, AI endpoints: 30/min, Communications: 20/min).

* Database WAL Mode:

SQLite running in Write-Ahead Logging (WAL) mode for maximum concurrency and zero database locking during concurrent reads/writes.

* Automated Database Backups:

Daily timestamped snapshots stored in encrypted, isolated storage with verified one-click restore scripts (backup_db.py).

STRICT CREDENTIAL ISOLATION

ZERO-CREDENTIALS POLICY: Production credentials, database files (*.db), secret keys, and active .env files must NEVER be committed to Git or packaged into team distribution archives. All deployments must use sanitized templates (.env.example).

SECTION 22

DATA IMPORT & EXPORT PORTABILITY

Seamless Data Portability:

To support data migration from legacy spreadsheets and enable executive analysis, KCT CRM includes a robust CSV data pipeline:

* Supported Entities:

Bulk import and export for Companies, Contacts, Leads, and Products/Services.

* Downloadable CSV Templates:

Standardized pre-formatted templates with exact column headers and sample data rows.

* Schema & Format Validation:

Ingestion engine verifies email syntax, telephone formats, required fields, and data types before processing.

* Duplicate Detection Engine:

Identifies existing records based on unique email addresses, company domains, or phone numbers.

* Interactive Import Preview:

Displays a pre-ingestion validation modal highlighting total rows, valid rows, and invalid rows.

* Granular Error Reporting:

Provides line-by-line downloadable error reports specifying exact validation failure reasons.

* One-Click Export:

Filtered views can be exported instantly to clean, UTF-8 encoded CSV files for accounting and executive review.

SECTION 23

BUSINESS AUTOMATION ENGINE (FUTURE V2)

Eliminating Repetitive Human Overhead:

The planned v2 Automation Engine introduces an event-driven trigger-action framework to orchestrate cross-departmental operations seamlessly:

Trigger Event	Automated Conditions	Orchestrated Action Chain
New Lead Captured	Inbound lead from Website or Ads; unassigned.	1. Auto-assign via round-robin rule -> 2. Create introductory follow-up -> 3. Alert owner via in-app toast.
Opportunity Marked Won	Sales stage changed to 'Won' with signed quotation.	1. Spawn delivery Project -> 2. Generate default milestones -> 3. Notify Project Manager and Finance.
Support Ticket Logged	Customer submits ticket via portal or email.	1. Apply SLA policy by priority -> 2. Route to on-call support tier -> 3. Initialize countdown timer.
Contract Expiring in 45 Days	Active contract reaches expiration window.	1. Spawn Renewal Opportunity -> 2. Create high-priority task for Account Owner -> 3. Send digest email.
Invoice Overdue by 7 Days	Invoice status = UNPAID and due date < today.	1. Flag Company credit status -> 2. Dispatch polite payment reminder -> 3. Alert Finance collection officer.

SECTION 24

NOTIFICATION CENTER & IN-APP ALERTS (FUTURE V2)

Real-Time Situational Awareness: The future Notification Center consolidates urgent operational alerts into a top-navigation drawer and instant desktop toasts:

- * **Lead Assignments:** Instant alert when a new prospect is routed to a sales representative.
- * **Overdue Follow-up Warnings:** Real-time reminders when a customer follow-up passes its scheduled deadline.
- * **Won Deal Celebrations:** Company-wide notification when a major opportunity closes.
- * **Project Delivery Delays:** Alerts when a delivery milestone slips past its scheduled completion date.
- * **Critical Bug Discovery:** Immediate high-priority dispatch to PM and lead developers when a P1 bug is filed.
- * **SLA Breach Warnings:** Yellow alert at 75% elapsed SLA time; Red alert upon contractual breach.
- * **Invoice Overdue Alerts:** Notification to account managers when a client exceeds payment terms.

SECTION 25

CUSTOMER LIFECYCLE & RELATIONSHIP CULTIVATION

1. Unknown Market	2. Identified Lead	3. Qualified Prospect	4. First-Time Customer	5. Active Client	6. Retention & Support	7. Renewal Client	8. Referral Partner
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Maximizing Customer Lifetime Value (LTV): The ultimate goal of KCT CRM is not merely closing one-off transactions, but cultivating long-term, multi-year institutional relationships:

- * **1. Discovery & Capture:** Transforming anonymous market interest into an identified lead with verified contact credentials.
- * **2. Value Alignment & Proof:** Demonstrating Kiwi Cloud Tech's technical expertise through counselling, customized demos, or architecture reviews.
- * **3. Commercial Commitment:** Negotiating transparent, fair quotations and executing legal contracts.
- * **4. Flawless Delivery:** Delivering on-time, bug-free software milestones and high-impact training curriculums.
- * **5. Proactive Support:** Resolving operational issues rapidly within contractual SLA windows.
- * **6. Retainer Renewal & Expansion:** Expanding account footprint from initial project into multi-year support retainers and additional service lines.
- * **7. Referral Advocacy:** Turning satisfied corporate clients into vocal brand advocates who refer new business.

SECTION 26

DOCUMENT MANAGEMENT & CLIENT VAULT (FUTURE V2)

Centralized Client Document Repository: The planned v2 Document Vault consolidates all corporate collateral directly under the master Company 360 profile:

Company Document Vault
■■■■ Master Services Agreements (MSAs) & Legal Non-Disclosure Agreements (NDAs)
■■■■ Statements of Work (SOWs) & Technical Proposal Specifications
■■■■ Formal Line-Item Quotations & Customer Purchase Orders (POs)
■■■■ Project Architecture Blueprints, Wireframes & Delivery Sign-off Certificates
■■■■ QA Test Execution Reports & Final Release Verification Sign-offs
■■■■ Customer Invoices, Receipts & Annual Tax Withholding Certificates

Key Document Capabilities: 1. Granular role-based access control (sales reps cannot access internal financial tax certificates), 2. Cryptographic SHA-256 checksums verifying file authenticity, 3. Automatic version control (SOW_v1, SOW_v2), and 4. Tamper-proof audit logging recording every document download and preview.

SECTION 27

AI COPILOT & AUGMENTED INTELLIGENCE (FUTURE V3)

AI as an Augmentation Layer, NOT a Replacement: In KCT CRM, Artificial Intelligence is architected strictly as an intelligence accelerator. AI suggests, summarizes, and drafts, but **never replaces human business judgment or transactional database integrity**:

AI Capability	Underlying Machine Learning Model	Practical Daily CRM Value
1. Lead Summarization	LLM Context Synthesizer	Summarizes long web inquiries and chat transcripts into a 3-bullet briefing.
2. Predictive Lead Scoring	Logistic Regression / Random Forest	Ranks incoming leads (1-100) based on domain authority, source, and budget.
3. Next-Action Recommendation	Sequence-to-Sequence Policy Model	Suggests optimal next outreach step based on successful historical deal paths.
4. Call Transcript Analysis	Speech-to-Text + Entity Extraction	Extracts action items, budget mentions, and competitor objections from recorded calls.
5. Contextual Email Drafting	Few-Shot Prompt Engineering	Generates personalized follow-up email drafts referencing previous discussion notes.
6. Company 360 Synthesis	Cross-Entity Rag Pipeline	Generates a 30-second executive dossier before an account executive joins a client call.
7. Support Ticket Summarization	Extractive / Abstractive Summarizer	Condenses a 20-message support thread into key symptoms and attempted fixes.
8. Response Recommendation	Vector Semantic Search on Solved Tickets	Suggests proven resolution steps to support agents based on historical tickets.

9. Predictive Sales Forecasting	Time-Series Revenue Regressor	Forecasts quarterly cashflow by analyzing rep velocity and deal slip patterns.
10. Churn Risk Warning	Survival Analysis / Anomaly Detection	Flags accounts showing declining ticket activity, overdue invoices, and unread emails.

HUMAN-IN-THE-LOOP MANDATE

CARDINAL AI ENGINEERING PRINCIPLE: AI models must never write directly to core transactional tables without explicit human-in-the-loop verification. All AI-generated outputs (drafts, scores, summaries) exist as advisory overlays.

SECTION 28

NATIVE MOBILE APPLICATION (FUTURE V3)

- Empowering Field Sales & Remote Delivery:** The planned v3 native mobile application (iOS & Android) will deliver on-the-go execution capabilities for team members in the field:
- * **One-Touch Click-to-Call:** Dial prospects directly from the mobile app with automatic call duration logging and prompt for outcome notes.
 - * **Voice-to-Text Note Dictation:** Sales reps dictate meeting notes immediately after client discussions, auto-transcribing into the activity feed.
 - * **Instant Push Notifications:** High-priority alerts for new inbound leads, overdue follow-ups, and P1 ticket escalations.
 - * **Offline Lead & Contact Access:** Local caching enabling account executives to review client dossiers even during poor network connectivity.
 - * **Mobile Pipeline Drag-and-Drop:** Rapidly update opportunity stages and close dates directly from smartphones.
 - * **Field Service Ticket Triage:** Support engineers can review, comment, and resolve critical incidents on the go.

SECTION 29

PRODUCT ROADMAP - VERSION 2.0 (OPERATIONS & SCALE)

Version 2.0 Strategic Focus: Transitioning from core operational stability to deep automation, live communication activation, and enriched customer interactions:

Functional Pillar	Planned V2 Features & Deliverables	Target Business Impact
1. Live Communications	Activation of live Hostinger SMTP/IMAP email sync, WhatsApp Cloud API, and cloud click-to-call.	Zero app switching; 100% centralized communication audit history.
2. Advanced Sales & Quotes	Branded PDF quote generation, multi-tier discount approval gates, digital e-signature.	Quote turnaround <10 min; elimination of pricing rogue discounts.
3. Contract Management	Automated contract renewal reminders (45-day trigger), master agreement versioning.	Zero surprise contract expirations; 25% higher retainer renewal rate.
4. Business Automation	Rule-based event trigger engine for auto-assignment, task generation, and SLA escalations.	30% reduction in manual administrative management tasks.
5. Notification Center	Real-time in-app notification drawer, desktop push toasts, sound badges for urgent alerts.	Instant team reaction to critical customer events and new leads.
6. Customer Portal	Secure client portal for viewing project milestones, downloading invoices, and filing tickets.	Enhanced client transparency; 40% reduction in status-update inquiries.
7. Document Vault	Encrypted document repository under Company 360 with access control and versioning.	Secure, organized client collateral management.

SECTION 30

PRODUCT ROADMAP - VERSION 3.0 (INTELLIGENCE & ECOSYSTEM)

Version 3.0 Strategic Focus: Infusing augmented intelligence, native mobile accessibility, and enterprise ecosystem integrations into the mature platform:

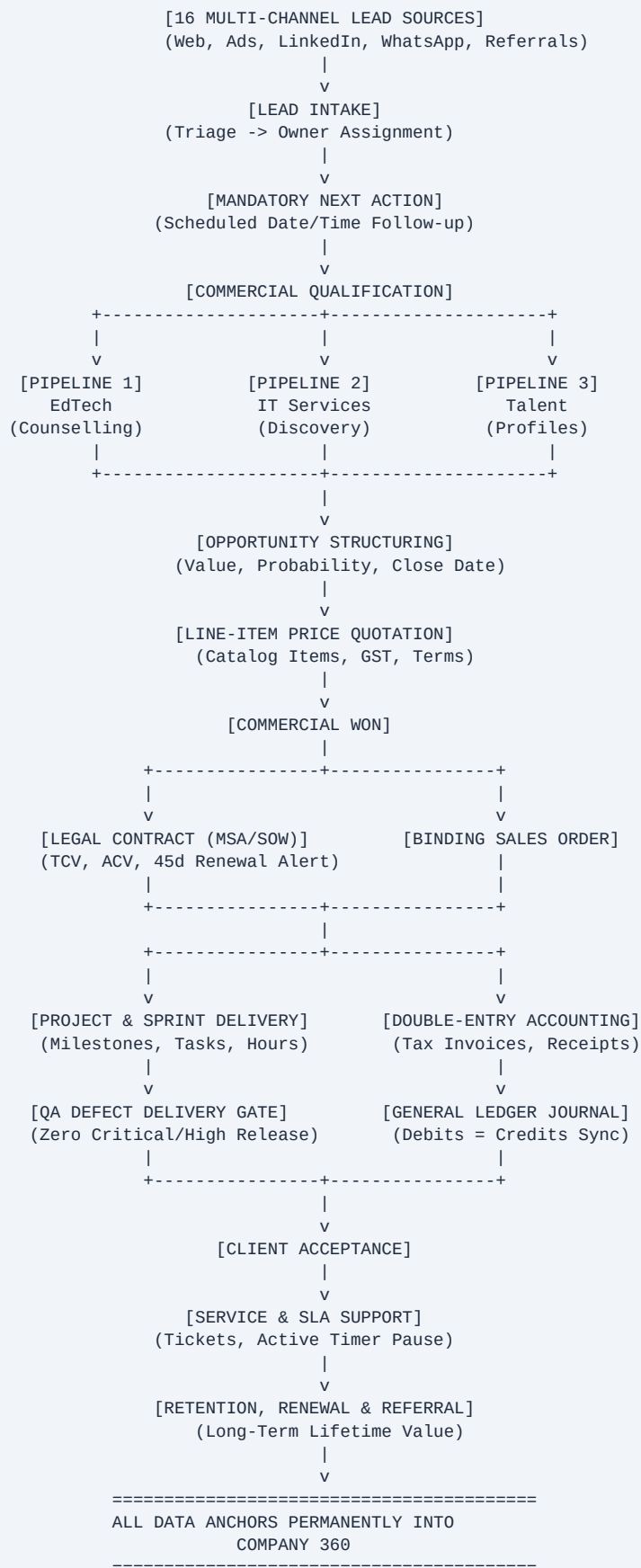
Next-Gen Pillar	Planned V3 Capabilities & Technical Scope	Strategic Advantage
1. AI Copilot Engine	Lead summarization, predictive lead scoring, automated draft generation, call transcription.	3x sales rep productivity; automated administrative summarization.
2. Native Mobile Apps	React Native / Flutter apps for iOS & Android with offline sync, call logging, push alerts.	Uninterrupted field sales execution and mobile incident response.
3. Predictive Analytics	Machine learning sales velocity forecasting, customer churn risk analysis, SLA predictive breach.	Proactive executive intervention before revenue or clients are lost.
4. Marketing Automation	Multi-stage automated email/WhatsApp drip sequences triggered by lead segment behavior.	Automated lead nurturing across the 16 marketing acquisition channels.
5. LMS Bridge (If Mandated)	One-way API synchronization linking completed EdTech corporate sales to external training systems.	Preserves CRM boundaries while servicing enterprise corporate training.
6. Enterprise ERP Sync	Bi-directional connector synchronizing general ledger journal entries with SAP / Tally / Zoho Books.	Seamless statutory accounting compliance for large enterprise audits.

PHASING DISCIPLINE
ROADMAP REALITY CHECK: V2 and V3 features must NOT be treated as prerequisites for current production operations. The current baseline platform is already fully production-capable for commercial sales, delivery, QA, support, and billing.

SECTION 31

END-TO-END MASTER BUSINESS ARCHITECTURE WORKFLOW

The Complete Unified Operational Lifecycle: The following master architecture illustrates how every subsystem and data transition interlocks seamlessly through the central **Company 360** hub:



SECTION 32

CURRENT STATE VS. PLANNED CAPABILITIES

Transparent Platform Status: To ensure flawless alignment between engineering, management, and operations, the following matrix delineates what is **Live & Functional** today versus what is **Intentionally Stubbed** or **Planned**:

Subsystem Area	Current Status in Codebase	Active Operational Capabilities	Target Future Scope
Authentication & RBAC	CURRENTLY LIVE	Argon2/bcrypt, JWT access/refresh, role permissions.	MFA / SSO Google OAuth (v2).
Company 360	CURRENTLY LIVE	Full relational master account linking contacts, deals, tickets.	Client-facing document vault (v2).
Leads & Follow-ups	CURRENTLY LIVE	16 sources, mandatory next-action date, Today's Work Center.	Automated lead scoring & AI summaries (v3).
Three Pipelines	CURRENTLY LIVE	EdTech, IT Services, and Talent isolated state machines.	Custom pipeline stage creator (v2).
Quotes & Orders	CURRENTLY LIVE	Line items, unit pricing, GST tax engine, Sales Orders.	Dynamic PDF generator & e-signatures (v2).
Project Delivery	CURRENTLY LIVE	Milestones, tasks, assignments, hours tracking.	Gantt charts & client milestone portal (v2).
QA & Defect Gate	CURRENTLY LIVE	Bug logging, severity, blocking gate on Critical/High bugs.	Automated CI/CD test suite integration (v3).
Service Desk & SLA	CURRENTLY LIVE	Tickets, status transitions, WAITING_FOR_CUSTOMER pause.	SLA automated breach escalation alerts (v2).
Accounting Foundation	CURRENTLY LIVE	Chart of accounts, invoices, payments, double-entry ledger.	Payment gateway auto-reconciliation (v2).
Email Communication	CURRENTLY DISABLED	Provider-neutral code abstraction (MAIL_PROVIDER=none).	Live Hostinger SMTP/IMAP sync (Phase 3).
Telephony Calling	CURRENTLY DISABLED	Provider-neutral code abstraction (TELEPHONY_ENABLED=false).	Live Twilio / Exotel cloud telephony (Phase 3).
WhatsApp Integration	CURRENTLY STUBBED	Webhook model and payload structures defined.	Live Meta Cloud API integration (Phase 3).
Audit Logging	CURRENTLY LIVE	Immutable audit_logs table tracking all system mutations.	Compliance audit visual log explorer (v2).
AI Copilot	CURRENTLY STUBBED	FastAPI AI router structure with mock response providers.	Live OpenAI / Gemini intelligence engine (v3).
Mobile Experience	CURRENTLY LIVE (WEB)	Fully responsive web UI supporting desktop and tablet.	Native iOS & Android mobile applications (v3).

SECTION 33

CORE ENGINEERING & DEVELOPMENT PRINCIPLES

The 15 Engineering Commandments: All engineering teams, technical architects, and contributors must strictly adhere to these immutable design principles when modifying or extending KCT CRM:

- * **1. Do Not Rebuild Existing Modules Unnecessarily:** Never discard proven, working code to chase minor refactorings. Extend existing patterns cleanly.
- * **2. Extend Existing Architecture Incrementally:** Build modular additions that hook into existing routers and services rather than introducing parallel frameworks.
- * **3. Do Not Duplicate Company / Contact Data:** The Company and Contact entities are the single source of truth. Never store parallel customer records in silos.
- * **4. Every Active Lead Must Have a Next Action:** Enforce next-action and follow-up date validation across all lead progression workflows.
- * **5. Keep the Three Business Pipelines Separate:** Never homogenize EdTech, IT Services, and Talent into a single generic status model. Respect distinct dynamics.
- * **6. Preserve Role-Based Access Control (RBAC):** Verify permission gates at both frontend navigation and backend FastAPI dependency levels on every endpoint.

- * **7. Preserve System Auditability:** Every financial modification, status shift, and permission change must record an immutable entry in `audit_logs`.
- * **8. Avoid Unnecessary Schema Changes:** Design data models thoughtfully. Migrations must be strictly additive and backward-compatible with existing data.
- * **9. Avoid Over-Engineering the MVP:** Solve real operational business problems first before architecting hyper-complex theoretical abstractions.
- * **10. Build Business Workflows Before Adding AI:** A rock-solid relational process with clear human accountability must precede any AI intelligence overlay.
- * **11. Integrations Must Use Provider Abstractions:** Code to abstract interfaces (e.g. `MailProvider`, `TelephonyGateway`) so providers can swap without core refactors.
- * **12. Do Not Expose Secrets or Credentials:** Never commit credentials, API tokens, database files, or active `.env` files to Git or distribution archives.
- * **13. Test Every Module Before Production:** Maintain unit and integration test coverage for all models, business routers, and permission boundaries.
- * **14. Preserve Backward Compatibility:** Ensure existing client contracts, invoices, and historic activities remain accessible across all upgrades.
- * **15. Strictly No Student / LMS Functionality:** Do not add student video players, LMS courseware, or student grading unless explicitly authorized by leadership.

SECTION 34

TEAM DEVELOPMENT LIFECYCLE & SAFETY PROTOCOL

Standardized Engineering SDLC Progression:

1. Requirement Review	2. Functional Design	3. Schema Review	4. Backend API	5. Frontend UI	6. Pytest Suite	7. Regression Test	8. UAT Sign-off	9. Documentation	10. Git & Deploy
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Mandatory Pre-Modification Checklist: Before any developer touches an existing module in `D:\kct\crm`, they must execute the following safety protocol:

- * **1. Understand Current Behavior:** Inspect active routers, schemas, and UI components before writing a single line of code.
- * **2. Identify Cross-Module Dependencies:** Check foreign key relationships and shared service utilities across Sales, Delivery, and Accounting.
- * **3. Avoid Breaking Existing Workflows:** Ensure enhancements do not invalidate existing client records, sales pipelines, or ticket histories.
- * **4. Execute Relevant Pytest Suites:** Run `pytest backend/tests/` to confirm zero regressions in existing business logic.
- * **5. Verify Database Schema Integrity:** Inspect Alembic migration histories; ensure SQLite constraints remain intact without orphan records.
- * **6. Validate Frontend Production Build:** Run `npm run build` inside `frontend/` to guarantee TypeScript compilation and clean routes.

SECTION 35

RECOMMENDED PHASED DEVELOPMENT PRIORITY

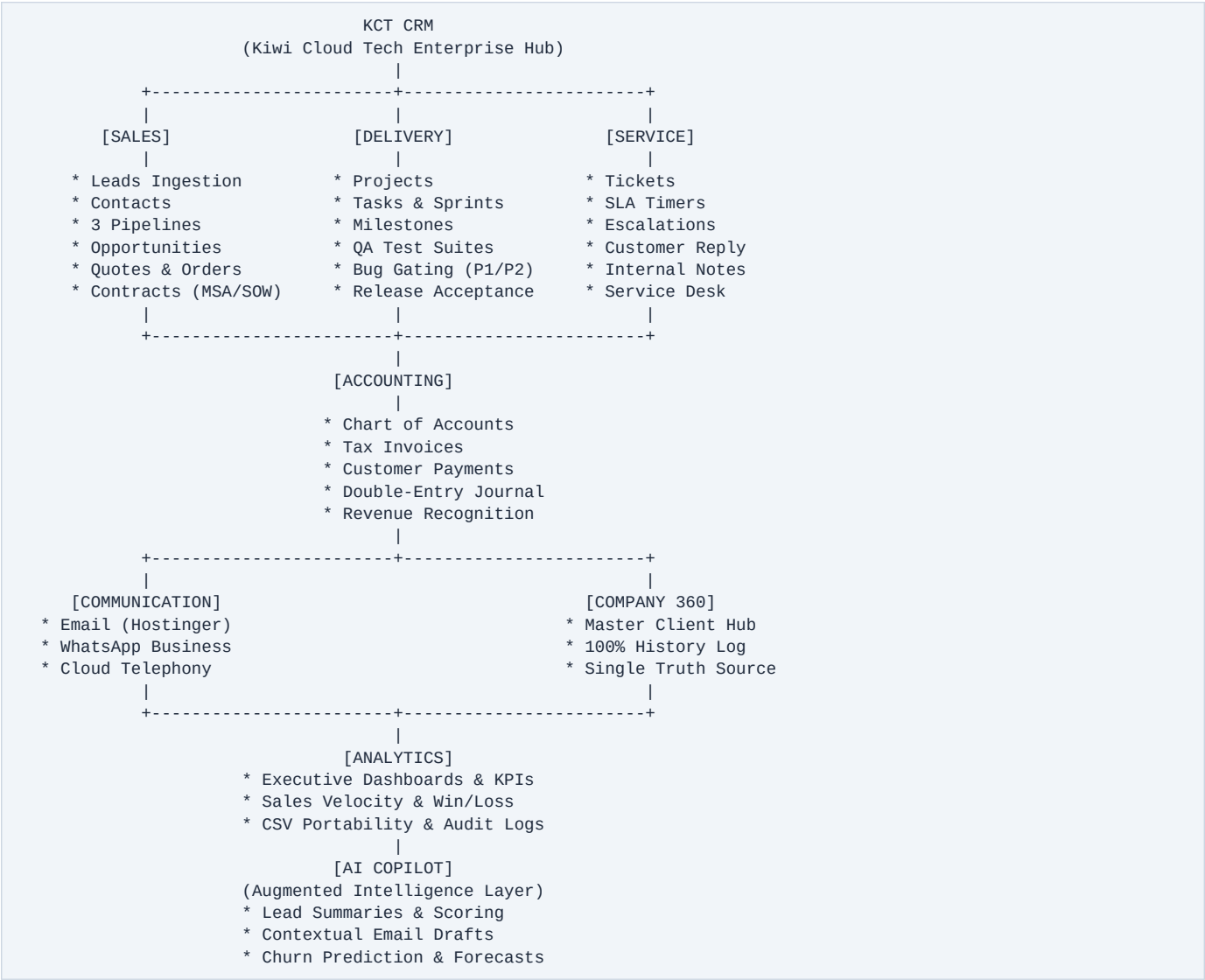
Strategic Master Development Roadmap: To maximize business velocity and maintain operational stability, engineering resources must execute projects in the following disciplined sequence:

Phase	Strategic Focus Area	Key Deliverables & Milestones	Priority
Phase 1	Production Deployment & Hardening	Dockerize production stack, enforce HTTPS, configure Nginx, verify WAL mode backups.	IMMEDIATE (P0)
Phase 2	Real Business Configuration	Populate Kiwi Cloud Tech production company settings, service catalog, roles, and initial users.	CRITICAL (P1)
Phase 3	Live Communications Activation	Connect Hostinger SMTP/IMAP email, WhatsApp Cloud API, and cloud telephony business lines.	HIGH (P2)
Phase 4	Automation & Real-time Notifications	Implement rule-based trigger engine, in-app notification drawer, and desktop alerts.	HIGH (P2)
Phase 5	Advanced Sales & Contract Governance	Dynamic branded PDF quotations, multi-level discount approvals, 45-day renewal alerts.	MEDIUM (P3)
Phase 6	Customer Portal & Document Vault	Client login portal for milestone tracking and ticket filing; encrypted company document vault.	MEDIUM (P3)
Phase 7	Advanced Executive Analytics	Cohort retention analytics, sales velocity modeling, customer churn risk analysis.	MEDIUM (P4)
Phase 8	AI Copilot Integration	LLM lead summarization, predictive lead scoring, automated draft generation, ticket triage.	LOW (P5)
Phase 9	Native Mobile Applications	Native iOS and Android mobile apps for field sales executives and remote service engineers.	FUTURE (P6)

SECTION 36

FINAL PLATFORM ARCHITECTURE OVERVIEW

The Complete Kiwi Cloud Tech CRM Architecture: The diagram below illustrates the comprehensive multi-tier component hierarchy linking all operational disciplines into one cohesive enterprise platform:



Concluding Architectural Summary: KCT CRM embodies a complete, resilient, and highly disciplined operational engine tailored directly to Kiwi Cloud Tech's multi-vertical commercial model. By enforcing quality release gates, mandatory follow-up discipline, multi-pipeline isolation, double-entry financial integrity, and strict role-based access control, the platform ensures long-term institutional scalability, enterprise credibility, and rapid commercial growth.